Top 10 GO Biological Process Gene Sets Ranked by Potency of Perturbation (Sorted by BMD 25% Relative Deviation at the 5th Percentile)

DTTID: 108-020-004-000-5
Study Number:108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Lung

Date: 09 Jun 2025
Time: 09:35:54 AM

Female

Category Name	Input Genes/Platform Genes in Gene Set	% Gene Set Coverage	BMD_{rd25} at the 5th Percentile (ppm)	$BMDL_{rd25}$ at 5th Percentile - $BMDU_{rd25}$ at 5th Percentile (ppm)	Genes with Changed Direction Up	Genes with Changed Direction Down
GO:0007059 chromosome segregation	9/41	22.0	208.6	55.4 - 1082.9	0	9
GO:0140694 non-membrane-bounded organelle assembly	6/59	10.2	236.8	65.6 - NC	0	6
GO:0007049 cell cycle	6/45	13.3	247.7	69.6 - NC	0	6
GO:0048568 embryonic organ development	5/47	10.6	264.4	73.4 - 2060.4	3	2
GO:0045931 positive regulation of mitotic cell cycle	3/51	5.9	288.7	135.6 - NC	0	3
GO:1901989 positive regulation of cell cycle phase transition	3/43	7.0	288.7	135.6 - NC	0	3
GO:1901990 regulation of mitotic cell cycle phase transition	7/94	7.5	288.7	135.6 - NC	1	6
GO:0051301 cell division	8/84	9.5	289.6	102.8 - NC	0	8
GO:1901987 regulation of cell cycle phase transition	11/116	9.5	293.1	122.5 - NC	1	10
GO:0050728 negative regulation of inflammatory response	5/51	9.8	293.3	143.2 - 448.6	2	3

rd25 = 25% relative deviation
NC = non-convergent

Active Genes:
GO:0007059: Ttk;Smc2;Nuf2;Mad2l1;Kif18b;Dlgap5;Ctcf;Cenpw;Cenpf
GO:0140694: Plk4;Dlgap5;Cirbp;Cenpw;Cdc20;Ccnb2
GO:0007049: Mybl1;Mki67;Cdca5;Kif18b;Cenpw;Cenpf
GO:0048568: Slc6a8;Junb;Hsd17b7;Gdnf;Cebpb
GO:0045931: Cdca5;Crebbp;Cdc20
GO:1901989: Cdca5;Crebbp;Cdc20
GO:1901990: Ttk;Mad2l1;Cdca5;Ier3;Crebbp;Cenpf;Cdc20
GO:0051301: Nuf2;Cdca5;Kif18b;Cenpw;Cdc20;Ccnb2;Ccnb2-ps2;Ccna2
GO:1901987: Ttk;Mapk14;Mad2l1;Cdca5;Ier3;Cry1;Crebbp;Cirbp;Cenpf;Cdc20;Apbb2
GO:0050728: Nr1d2;Ier3;Hgf;Enpp3;Ahr

Annotations
Official active gene symbols from the Rat Genome Database are shown in the "Active Genes" section above. Definitions of Gene Ontology terms were provided by the Gene Ontology Resource (http://geneontology.org/) and are included below.
GO process description version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

GO Biological Process Definitions:
GO:0007059 chromosome segregation: The process in which genetic material, in the form of chromosomes, is organized into specific structures and then physically separated and apportioned to two or more sets. In eukaryotes, chromosome segregation begins with the condensation of chromosomes, includes chromosome separation, and ends when chromosomes have completed movement to the spindle poles.
GO:0140694 non-membrane-bounded organelle assembly: This biological process involves the aggregation, arrangement, and bonding together of a set of components to form a non-membrane-bounded organelle. Unlike traditional organelles enclosed by lipid membranes (e.g., nucleus, mitochondria), membraneless organelles are dynamic structures formed through processes like liquid-liquid phase separation. Examples include the nucleolus, stress granules, and P bodies. These organelles play crucial roles in various cellular functions, such as RNA processing, signal transduction, and stress responses. Their assembly is essential for maintaining cellular organization and function.
GO:0007049 cell cycle: The progression of biochemical and morphological phases and events that occur in a cell during successive cell replication or nuclear replication events. Canonically, the cell cycle comprises the replication and segregation of genetic material followed by the division of the cell, but in endocycles or syncytial cells nuclear replication or nuclear division may not be followed by cell division.

DTTID: 108-020-004-000-5
Study Number:108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Lung

Date: 09 Jun 2025
Time: 09:35:54 AM

GO:0048568 embryonic organ development: Development, taking place during the embryonic phase, of a tissue or tissues that work together to perform a specific function or functions. Development pertains to the process whose specific outcome is the progression of a structure over time, from its formation to the mature structure. Organs are commonly observed as visibly distinct structures, but may also exist as loosely associated clusters of cells that work together to perform a specific function or functions.

GO:0045931 positive regulation of mitotic cell cycle: Any process that activates or increases the rate or extent of progression through the mitotic cell cycle.

GO:1901989 positive regulation of cell cycle phase transition: Any process that activates or increases the frequency, rate or extent of cell cycle phase transition.

GO:1901990 regulation of mitotic cell cycle phase transition: Any process that modulates the frequency, rate or extent of mitotic cell cycle phase transition.

GO:0051301 cell division: The process resulting in division and partitioning of components of a cell to form more cells; may or may not be accompanied by the physical separation of a cell into distinct, individually membrane-bounded daughter cells.

GO:1901987 regulation of cell cycle phase transition: Any process that modulates the frequency, rate or extent of cell cycle phase transition.

GO:0050728 negative regulation of inflammatory response: Any process that stops, prevents, or reduces the frequency, rate or extent of the inflammatory response.

** END OF REPORT **